Agentic AI: Autonomous UI & Microservices Evolution

Seamlessly evolving micro -frontends and microservices in cloud platforms.

Problem Context: The Dual Edge of Modularity

Microservices Benefits

- Independent deployment
- Scalability & resilience
- Team autonomy

Micro-Frontend Benefits

- Independent UI delivery
- Mixed frameworks (React, Angular)
- Faster feature rollout

New Challenges Emerge

- API changes break UI
- Version drift
- Manual coordination overhead
- Slow reaction to runtime issues

Why This Is Hard at Scale

In complex enterprise environments, the challenges multiply.

1

Organizational Complexity

- Dozens of teams
- Multiple release cycles
- Distributed ownership

2

Compounding Issues

- Frequent UI regressions
- API contract mismatches
- Increased incidents
- Hindered innovation

The key challenge: maintaining continuous alignment between backend services and frontend UIs.

What Is Agentic AI?

Software agents that can observe, decide, and act toward a defined goal.

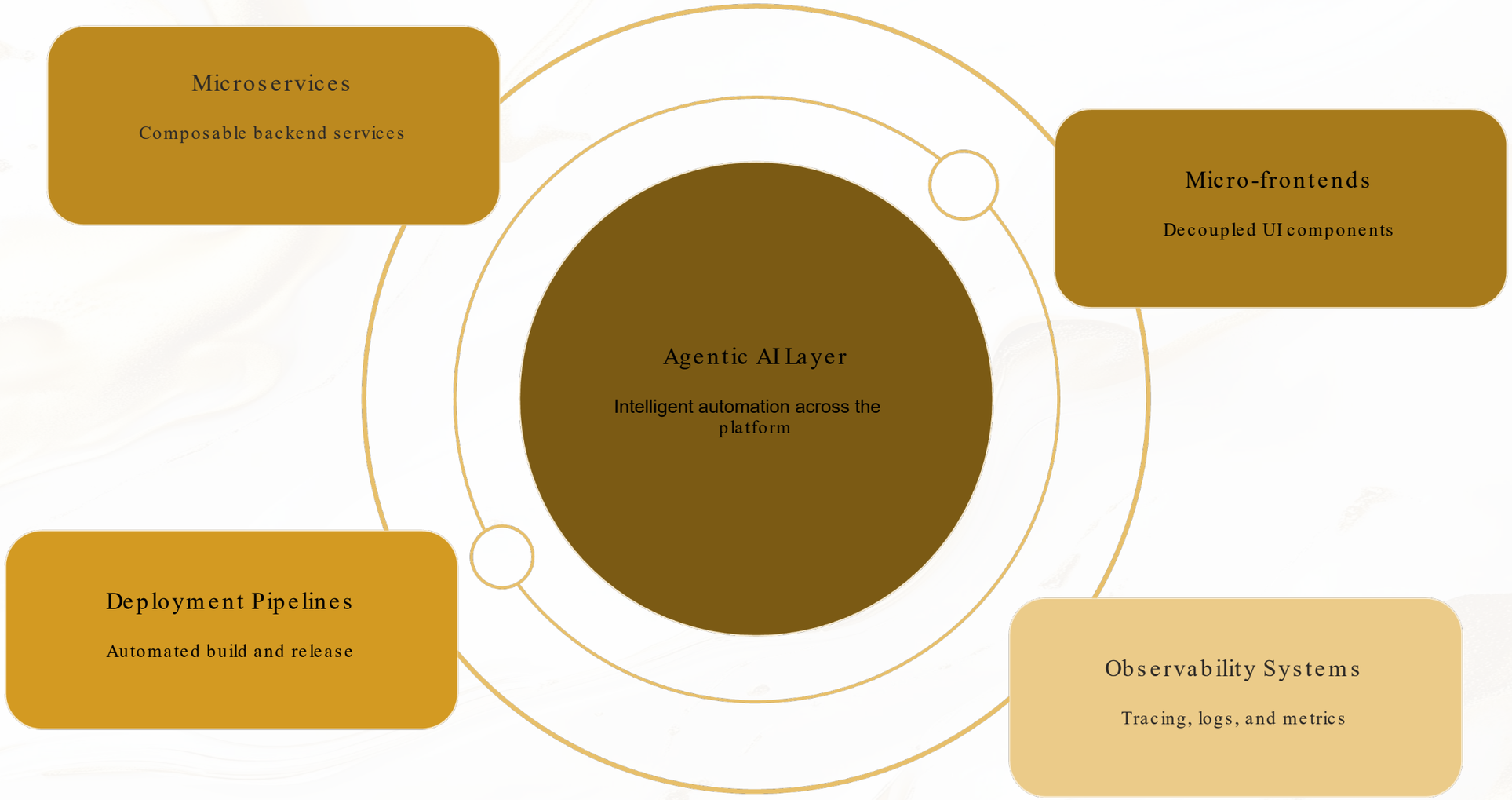
Observe
Metrics, logs, API changes, UI errors

Decide
Determine necessary actions

Act
Trigger workflows, propose changes, automate fixes

Agentic AI assists operations and coordination, it does not replace developers.

Agentic AI in the Platform Architecture



Agentic AI functions as an intelligent automation layer, integrating across the entire ecosystem.

Use Case: API Change Impact

Traditional Flow

- Backend API updates field
- UI breaks unexpectedly
- Users report the issue
- Team investigates root cause
- Manual fix and redeploy

With Agentic AI Assistance

- Detects contract change
- Identifies affected UI components
- Triggers compatibility checks
- Proposes update or rollback
- Deploys safely with approval gates

Micro-Frontend Coordination

Bridging React and Angular frontends for seamless operation.

Agent Capabilities

- Validates shared UI contracts
- Detects dependency conflicts
- Monitors performance regressions
- Coordinates version compatibility

Results in:

- Fewer UI outages
- Faster team alignment
- Reduced manual validation

Progressive Delivery with Guardrails

Agentic AI enhances safe, controlled rollouts.

Gradual Rollout

Leverages canary releases, feature flags, blue/green deployments.

Real-time Monitoring

Monitors error rates and latency continuously.

Automated Response

Automatically pauses or rolls back deployments on threshold breach.

Proactive Alerts

Notifies teams of critical issues.

This approach ensures risk is consistently controlled and managed.

Safety and Control Principles

Critical boundaries for responsible automation.

Human Oversight

Humans approve all critical changes.

Policy-Driven

Policies define what agents can modify.

Auditable Actions

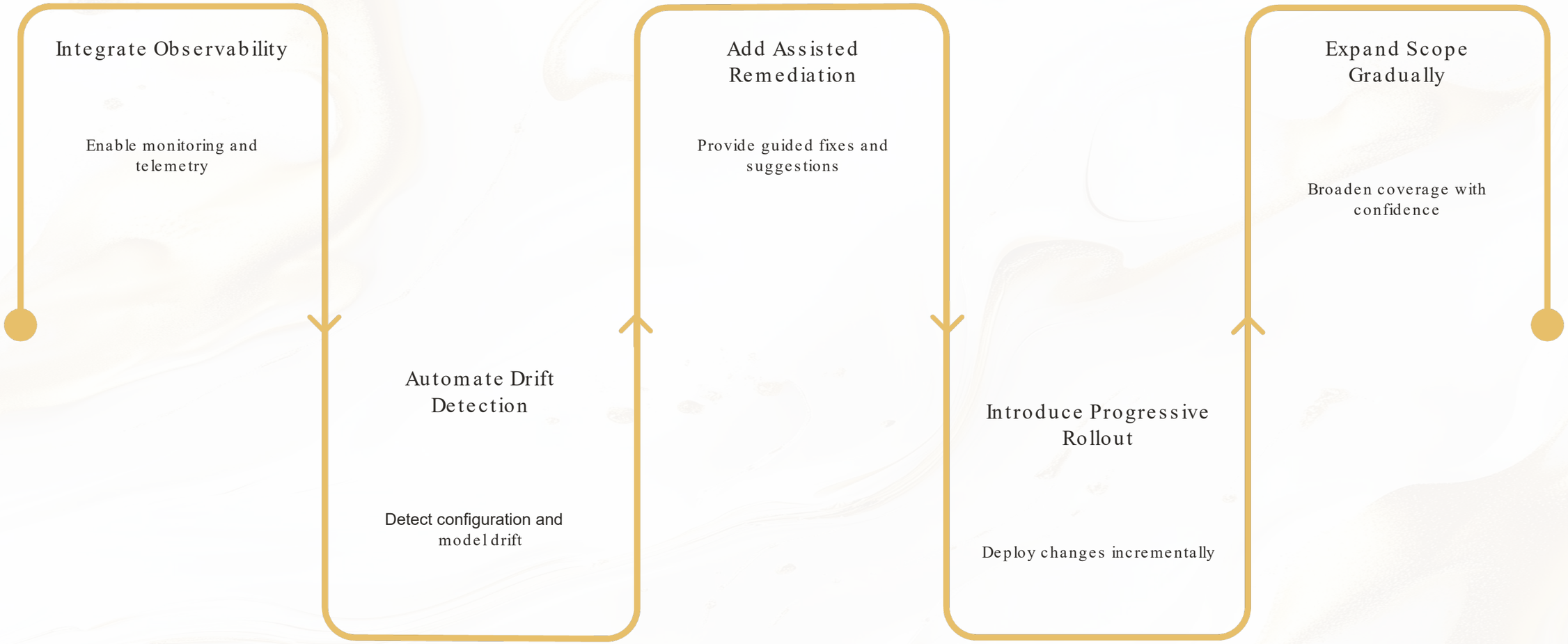
All agent actions are logged and auditable.

No Direct Code Writes

No production code writes without review.

Goal: Automation with accountability for secure and compliant operations.

Practical Adoption Path: Start Small



An incremental approach mitigates risk and builds confidence in Agentic AI capabilities.